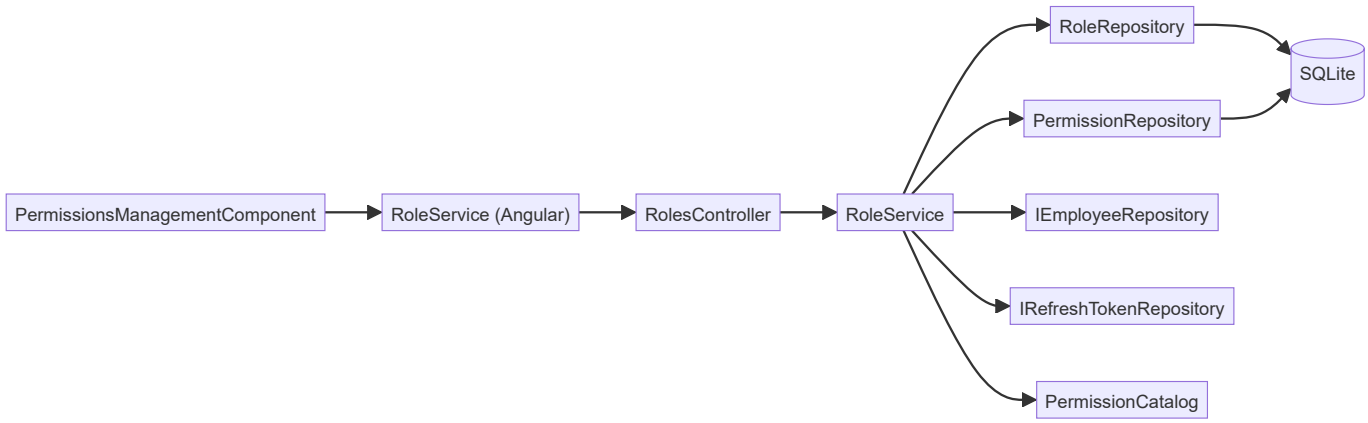
Role Configuration – Technical Documentation

> **Project:** TaskFlow – Intelligent Task Allocation & Tracking System **Parent Module:** Settings · **Child Module:** 08.3 Role Configuration **Version:** 1.0 · **Last Updated:** 31-Oct-2026 **Backend:** ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core) **Author:** Jojo George (jojogeorge3344@gmail.com) © 2026 Jojo George. All rights reserved.

1. Technical Overview

Role Configuration is `RolesController` (class-level `roles.manage`) over `RoleService`, `RoleRepository` and `PermissionRepository`. It manages custom `Roles` and their `RolePermissions` grants against the fixed permission catalogue in `Common/Security/PermissionCatalog.cs`. System roles and the `Permissions` table are protected by SQLite triggers. A permission change on a role signs out its holders: their security stamp is rotated and their refresh tokens are revoked (through `IEmployeeRepository` and `IRefreshTokenRepository`). The catalogue endpoints for create/update/delete always refuse.

2. Architecture Flow



3. Backend File Structure

Layer	Path	Files
Controller	`TaskFlow.Api/Controllers/Settings/`	`RolesController.cs`
Service	`TaskFlow.Api/Services/Settings/RoleConfiguration/`	`IRoleService.cs`, `RoleService.cs`
Repository	`TaskFlow.Api/Repositories/Settings/RoleConfiguration/Roles/`	`IRoleRepository.cs`, `RoleRepository.cs`, `RoleSummary.cs`
Repository	`TaskFlow.Api/Repositories/Settings/RoleConfiguration/Permissions/`	`IPermissionRepository.cs`, `PermissionRepository.cs`
DTOs	`TaskFlow.Api/Dtos/Settings/RoleConfiguration/`	`RoleDtos.cs` (`PermissionDto`, `RoleDto`, `RoleOptionDto`, `SaveRoleRequest`)
Validators	–	None. Name and permission rules are in `RoleService`

Layer	Path	Files
Entities	`TaskFlow.Api/Models/Entities/`	`Role.cs`, `Permission.cs`, `RolePermission.cs`
Security	`TaskFlow.Api/Common/Security/`	`PermissionKeys.cs`, `PermissionCatalog.cs`, `EffectivePermissions.cs`, `SystemRoles.cs`
Exceptions	`TaskFlow.Api/Common/Exceptions/`	`RoleException.cs` (`RoleFailure`)
Seeders	`TaskFlow.Api/Persistence/Seeders/`	`AccessControlSeeder.cs`
Migrations	`TaskFlow.Api/Persistence/Migrations/`	`20261019133820_AddRolesAndPermissions.cs` (triggers)

4. Frontend File Structure

Type	Path	Purpose
Page component	`src/app/modules/settings/role-configuration/pages/permissions-management.component.{ts,html}`, `permissions-management.model.ts`	`PermissionsManagementComponent` (tree + matrix)
API service	`src/app/modules/settings/role-configuration/services/role.service.ts`	`RoleService.list()/permissions()/create()/update()/delete()/assignable()`
Models	`src/app/modules/settings/role-configuration/models/role.model.ts`	`AccessRole`, `PermissionInfo`, `SaveRoleRequest`, `SYSTEM_ROLE_IDS`
Utils	`src/app/modules/settings/role-configuration/utils/permission-tree.ts`	`buildTree`, `toggleKey` (auto-ticks `requires`), `setScope`, `setMany`, `restrictForCaller` (`RESTRICTED_KEYS`), `isSelectable`
Route / Guard	`src/app/modules/settings/settings.routes.ts` (`settings/permissions`), `src/app/shared/auth/auth.guard.ts`	`permissionGuard` with `RolesManage`
Menu	`src/app/layouts/nav-items.ts`	"Role Configuration" item when `RolesManage` is held

5. API & Logic

5.1 API Endpoints

All routes are under `[Route("api/v1")]` with class-level `[RequirePermission(roles.manage)]`.

#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
1	GET	`/api/v1/roles`	All roles (system first). Without `roles.view.all` only the caller's own role is returned	`roles.manage` (+ claim check `roles.view.all`)	–	`IReadOnlyList<RoleDto>`
2	GET	`/api/v1/roles/{id:int}`	One role	`roles.manage`	–	`RoleDto`

#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
3	POST	`/api/v1/roles`	Create custom role	`roles.manage` + claim `roles.modify.access` (else `Forbid()`)	`SaveRoleRequest`	`201 Created` + `RoleDto`
4	PUT	`/api/v1/roles/{id:int}`	Rename custom role and replace its permissions	`roles.manage` + claim `roles.modify.access`	`SaveRoleRequest`	`RoleDto`
5	DELETE	`/api/v1/roles/{id:int}`	Delete custom role nobody holds	`roles.manage` + claim `roles.modify.access`	–	`204 No Content`
6	GET	`/api/v1/permissions`	The fixed permission catalogue in display order	`roles.manage`	–	`IReadOnlyList<PermissionDto>`
7	POST	`/api/v1/permissions`	Always refused (`CatalogIsFixed`)	`roles.manage`	–	`403` ProblemDetails
8	PUT	`/api/v1/permissions/{id:int}`	Always refused	`roles.manage`	–	`403` ProblemDetails
9	DELETE	`/api/v1/permissions/{id:int}`	Always refused	`roles.manage`	–	`403` ProblemDetails

Related (Management module): `GET /api/v1/users/roles` in `EmployeesController` returns `RoleOptionDto` for the employee form. It does not include permissions.

5.2 Request / Response Samples

```
// POST /api/v1/roles
{ "name": "Team Lead", "permissionKeys": ["tasks.view.all", "tasks.create", "workload.view"] }
```

```
// 201 Created
{ "id": 4, "name": "Team Lead", "isSystem": false, "userCount": 0, "permissionKeys":
["tasks.create", "tasks.view.all", "workload.view"] }
```

```
// GET /api/v1/permissions (one entry)
{ "id": 52, "key": "settings.ai.edit", "name": "Edit AI provider settings", "description":
"Configure AI providers, keys and models.", "group": "Settings", "isAssignable": true,
"module": "Settings", "screen": "TFA Settings", "section": "AI providers", "action": "Edit",
"requires": "settings.ai.view", "scopeGroup": null, "scopeRank": 0, "alwaysGranted": false }
```

> The `id` values in the samples are illustrative. Always-on entries return `id: 0`.

5.3 Service Layer Logic

Method: `RoleService.GetRolesAsync()`

- `RoleRepository.GetSummariesAsync()` returns the roles. `PermissionKeys` = `EffectivePermissions.Visible(EffectivePermissions.Expand(stored))` : legacy keys are expanded to their finer children, lower scope levels are added, and keys not in the catalogue are dropped.

Method: `RoleService.GetPermissionsAsync()`

- Maps `PermissionCatalog.All` to `PermissionDto`, taking the `Id` from `PermissionRepository.GetIdsByKeyAsync()`. The hierarchy comes from code, not from the table.

Method: `RoleService.CreateAsync(SaveRoleRequest)`

1. `ValidateName` : trim. Empty gives `RoleFailure.Invalid` "Enter a name for the role.". Over 100 gives "The role name can be at most 100 characters.".
2. `ResolvePermissionsAsync` :
    - trim and distinct the keys
    - expand legacy keys through `EffectivePermissions.LegacyChildren`
    - drop `AlwaysGranted` keys
    - unknown keys give "Unknown permission: …"
    - a missing `Requires` gives ""X" also needs "Y"."
    - an `IsAssignable = false` key (taken from the catalogue) gives "… can only be held by the Administrator role and cannot be given to another role."
3. `EnsureNameFreeAsync` : `NameTakenAsync` (NOCASE column) gives `RoleFailure.Duplicate` "A role named "…" already exists.".
4. Creates `Role { IsSystem = false, CreatedAt = now }` with its `RolePermission` rows. `SaveAsync` maps `UniqueConstraintException` to "A role with that name already exists." (409).

Method: `RoleService.UpdateAsync(int id, SaveRoleRequest)`

1. `LoadCustomRoleAsync` : not found gives 404 "That role does not exist.". `IsSystem` gives `RoleFailure.ReadOnly` (403) "{name} is a system role and is read-only: it cannot be changed or deleted.".
2. Validates the name and permissions as in Create, then diffs the grants (removes the unwanted ones, adds the missing ones).
3. If the grants changed, `EndSessionsOfRoleHoldersAsync` sets a new `SecurityStamp` (`Guid.NewGuid().ToString("N")`) on every holder and sets `RevokedAt = now` on their active refresh tokens.
4. Saves and returns the role.

Method: `RoleService.DeleteAsync(int id)`

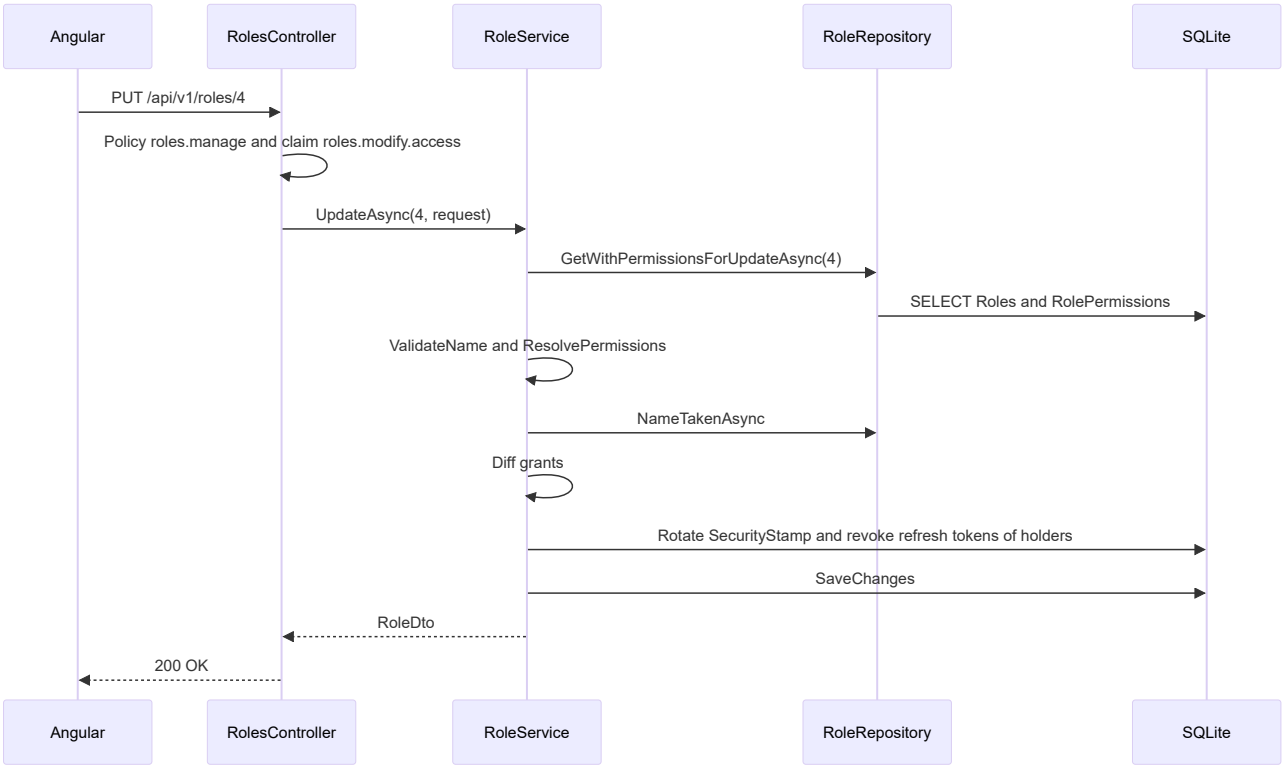
1. `LoadCustomRoleAsync` (same rules as above).
2. `IEmployeeRepository.CountHoldingRoleAsync(id)` > 0 gives `RoleFailure.InUse` (409) "1 user still holds this role. Move them to another role first." or "{n} users still hold this role. Move them to another role first.".
3. Removes the role. The `RolePermissions` rows are deleted by cascade.

5.4 Repository Layer

Method	Description	Query Notes
`RoleRepository.GetSummariesAsync()`	Roles with user count and stored keys	`AsNoTracking` , `OrderByDescending(IsSystem).ThenBy(Id)` . `UserCount` uses `Db.Users.IgnoreQueryFilters()` , so soft-deleted accounts are counted

Method	Description	Query Notes
`RoleRepository.GetWithPermissionsForUpdateAsync(id)`	Tracked role with grants	`Include(Permissions)`
`RoleRepository.NameTakenAsync(name, exceptRoleId)`	Name collision check	`AnyAsync`. The column collation is `NOCASE`
`RoleRepository.GetPermissionKeysByRoleNameAsync(roleName)`	Keys held by a role name	Used by `TaskService` (Monitoring)
`PermissionRepository.GetIdsByKeyAsync()`	Key to Id map	`AsNoTracking`
`PermissionRepository.GetByKeysAsync(keys)`	Tracked permission rows to grant	`Where(keys.Contains(Key))`
`IEmployeeRepository.CountHoldingRoleAsync` / `GetHoldingRoleForUpdateAsync`	Holders of a role	Management module
`IRefreshTokenRepository.GetActiveForUsersForUpdateAsync`	Active refresh tokens of the holders	Auth module

5.5 Sequence Diagram



6. Database Design

Table	Column	Type	Constraints	Description
`Roles`	`Id`	int	PK	1 Administrator, 2 Manager, 3 Employee (system)
`Roles`	`Name`	string(100)	Not Null, Unique, Collation `NOCASE`	Role name
`Roles`	`IsSystem`	bool	Not Null	System roles are read-only
`Roles`	`CreatedAt`	DateTime	Not Null	UTC
`Permissions`	`Id`	int	PK	Row id
`Permissions`	`Key`	string(100)	Not Null, Unique	e.g. `settings.ai.edit`
`Permissions`	`Name`	string(200)	Not Null	Display name
`Permissions`	`Description`	string(500)	Not Null	Description
`Permissions`	`Group`	string(100)	Not Null	Catalogue module
`Permissions`	`IsAssignable`	bool	Not Null	False = Administrator only
`RolePermissions`	`RoleId`	int	PK (composite), FK	Role
`RolePermissions`	`PermissionId`	int	PK (composite), FK	Permission

Also used (owned by other modules): `Users.RoleId` and `Users.SecurityStamp` (Management), `RefreshTokens.RevokedAt` (Authentication).

Relationships

- `RolePermissions.RoleId → Roles.Id` (FK, Cascade)
- `RolePermissions.PermissionId → Permissions.Id` (FK, Restrict)
- `Users.RoleId → Roles.Id` (FK, Restrict)

Database triggers (migration `20261019133820_AddRolesAndPermissions`):

- `TR_Roles_SystemNoUpdate` / `TR_Roles_SystemNoDelete` – "System roles are read-only." / "System roles cannot be deleted."
- `TR_Permissions_NoUpdate` / `TR_Permissions_NoDelete` – "The permission catalog is read-only."
- `TR_RolePermissions_SystemNoUpdate` / `TR_RolePermissions_SystemNoDelete` – "System roles are read-only." (inserts are allowed, so the seeder can add new grants)

Seed data: `AccessControlSeeder` creates the system roles and missing permissions. It grants every stored permission to the Administrator on every start. Manager and Employee get their `GrantedTo*` defaults only when a permission is first created.

7. Security & Authorization

- **Authorization:** class-level `[RequirePermission(roles.manage)]` on `RolesController`.
- **In-action claim checks:** `RolesController` checks `User.HasClaim("permission", roles.view.all)` and `roles.modify.access` inside the actions.
- **Data scope:** Role listing is narrowed to the caller's own role without `roles.view.all`.
- **Frontend:**
  - `permissionGuard` on `settings/permissions` and the menu item in `buildNavigation()`.
  - In-component checks `canViewAllRoles` and `canModifyAccess`.
  - `restrictForCaller()` marks `roles.administrator.assign` as not assignable for callers who do not hold it.

Permission Key	Endpoint(s)	UI Element Controlled
`roles.manage`	All `/api/v1/roles*` and `/api/v1/permissions*`	Menu "Role Configuration", route `/settings/permissions`
`roles.view.all`	`GET /api/v1/roles` (filter)	Role list shows all roles, not only one's own
`roles.modify.access`	`POST`, `PUT`, `DELETE /api/v1/roles`	"New role", editable name/tree, Copy from, Delete, Save bar
`roles.administrator.assign`	(Management `EmployeesController`)	Whether the Administrator-only entry is offered in the tree

8. Validations & Error Handling

Rule	Layer	HTTP Code	Error Message
Role name empty	`RoleService`	400	Enter a name for the role.
Role name > 100	`RoleService`	400	The role name can be at most 100 characters.
Unknown permission key	`RoleService`	400	Unknown permission: {keys}.
Required permission missing	`RoleService`	400	"{name}" also needs "{needed}".
Non-assignable permission	`RoleService`	400	{names} can only be held by the Administrator role and cannot be given to another role.
Duplicate role name	`RoleService`	409	A role named "{name}" already exists. / A role with that name already exists.
System role change/delete	`RoleService`	403	{name} is a system role and is read-only: it cannot be changed or deleted.
Role in use	`RoleService`	409	1 user still holds this role. Move them to another role first.
Role not found	`RoleService`	404	That role does not exist.
Catalogue mutation	`RolesController.CatalogIsFixed`	403	The permission list is fixed and cannot be changed.
Create/update/delete role without `roles.modify.access`	`RolesController` (`Forbid()`)	403	(empty body)

Errors from `RoleService` are `RoleException` (`RoleFailure`) and are mapped to ProblemDetails by `GlobalExceptionHandler`. Shared rules are in the parent document.

9. Integrations / Side Effects

- **Sessions:** A permission change on a custom role rotates `Users.SecurityStamp` and revokes the active `RefreshTokens` of every holder.
- **Authorization everywhere:** `SessionValidation` refreshes permission claims on each request, so grants saved here apply on the next call in every module.
- **Monitoring:** `TaskService` reads `GetPermissionKeysByRoleNameAsync`.
- **AI Assistant:** knowledge page `Services/Assistant/Knowledge/change-role.md`.
- **Activity Log:** no activity row is written for role changes.

10. Configuration

Not applicable: Role Configuration has no `appsettings.json` keys. The permission catalogue is defined in code (`PermissionCatalog.cs`).

11. Testing

Test Type	File / Location	Covers
Unit	`tests/TaskFlow.UnitTests/Settings/RoleConfiguration/PermissionCatalogTests.cs`	Unique keys, each with a place in the hierarchy; module and submodule layout; references and scope groups point at real entries; old keys are kept and expand to their finer keys; always-on entries are never stored; only the Administrator holds the Administrator-only key; the frontend mirrors the stored keys
Unit	`tests/TaskFlow.UnitTests/Persistence/RolesMigrationTests.cs`	The upgrade moves old roles onto system roles and ends sessions; triggers block changes to system roles, their grants and catalogue rows but allow inserts and custom-role changes; role names are unique ignoring case; the seeder gives default grants, gives the Administrator new entries and is idempotent
Unit	`tests/TaskFlow.UnitTests/Monitoring/TaskManagement/Tasks/CustomRolePermissionTests.cs`, `Monitoring/EmployeeWorkload/CustomRoleWorkloadTests.cs`	In other modules: the view-all tasks key decides which tasks a custom role sees; create, edit, reassign and recipients need their own keys; a removed key stops on the next call; workload lists custom-role users without view-all
Integration	`tests/TaskFlow.IntegrationTests/Settings/RoleConfiguration/RolesEndpointsTests.cs`	Administrator lists roles and catalogue and does custom-role CRUD; non-administrators, custom roles and anonymous callers are refused; system roles and the catalogue cannot be changed; role management can be delegated; permission changes end the role's sessions but a rename does not; duplicate, system or blank names, unknown permissions and the Administrator-only key are rejected; a role in use cannot be deleted; users can be moved to a custom role
Integration	`tests/TaskFlow.IntegrationTests/Settings/RoleConfiguration/RoleAssignmentSafeguardTests.cs`	A role that can assign roles but not manage them cannot create, promote or demote an Administrator; an Administrator can; an employee update never changes the password; the employee form's role list needs assign-roles only
Integration	`tests/TaskFlow.IntegrationTests/Common/ModuleBasedPermissionEndpointsTests.cs`	Role-saving cases: the catalogue endpoint returns each entry's place in the hierarchy; saving a role needs the View key its actions depend on; old keys are stored as their finer keys and always-on entries are ignored; a copy of a system role can be edited without changing the original

Test Type	File / Location	Covers
Frontend unit	`src/app/modules/settings/role-configuration/pages/permissions-management.component.spec.ts`	Role list, search and load error; system roles read-only with no save, delete or rename; ticks and counts what a role holds; View/action and dashboard-scope rules; module, screen and select-all toggles; search and filters; compare and copy roles; create, save and delete with confirmations; unsaved-edit guards; narrow-screen layout
Frontend unit	`src/app/modules/settings/role-configuration/utils/permission-tree.spec.ts`	Builds the tree in catalogue order with each module once; scope levels as one feature; View/action dependency; group toggles; search; counts changes; `restrictForCaller` locks only the Administrator-only key for a delegated user

12. Known Limitations / Gaps found

- Changing only a custom role's name does not sign its holders out; only a change to its permissions does.
- **Matrix vs code – `roles.administrator.assign`:** the matrix lists Custom role as "CONFIGURABLE", but the catalogue sets `IsAssignable: false`, and `RoleService.ResolvePermissionsAsync` refuses it for any custom role. The correct value is Denied.
- **Matrix vs code – `roles.manage` assignable to custom roles:** the catalogue has `IsAssignable = true`, while the `RolesController` / `RoleService` frontend comments say "Administrator only" (audit item 1). A custom role holding `roles.manage` + `roles.modify.access` can create a role with `employees.role.assign`.
- **`GET /api/v1/roles/{id}` ignores `roles.view.all`:** a holder of `roles.manage` without `roles.view.all` gets only their own role from `GET /roles`, but can still read any role by id.
- **"Duplicate as custom role"** is shown for system roles without `roles.modify.access`. The new draft is then not editable and cannot be saved.
- **Empty 403 bodies:** `Forbid()` in `RolesController` returns no ProblemDetails, so the UI shows only the fallback ("Could not save the role.").
- **Stale comments:** `RoleService` (Angular) and `PermissionsManagementComponent` say "Administrator only", but access is permission-based.

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